

Performance Bound Analysis of Compressed Sensing for Airborne Random PRI Radar

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Abstract—Airborne pulse-Doppler radar with random pulse repetition interval (PRI) waveforms resolves range and Doppler ambiguities but suffers from extensive clutter sidelobes under matched filtering. Compressed sensing (CS) has been introduced as a nonlinear reconstruction technique to suppress these sidelobes. However, its performance hinges on scenario sparsity and the number of measurements, which are not guaranteed in airborne scenarios with widely spread clutter. This paper analyzes the performance bounds of CS in random PRI radar using phase transition theory. We first verify that the random PRI measurement matrix exhibits phase transition behavior identical to that of Gaussian random matrices. Based on this universality, we derive the theoretical boundary for exact reconstruction by quantifying the clutter sparsity from radar geometry. Numerical simulations confirm that this theoretical curve effectively predicts the performance boundaries in both noiseless and noisy environments, providing rigorous guidelines for waveform design.

Index Terms—Random PRI, compressed sensing, performance bounds, phase transition.

I. INTRODUCTION

Airborne pulse-Doppler radar is crucial for target detection [1]–[3], yet its performance is often degraded by range-Doppler ambiguities and strong ground clutter [4], [5]. To resolve ambiguities, random PRI waveforms have been proposed to break the pulse train periodicity [6]–[8]. However, this randomness introduces a significant drawback: under conventional matched filtering, ground clutter manifests as high sidelobes that may mask weak targets [9]. This creates a pressing need for advanced processing techniques that achieve both ambiguity resolution and clutter suppression.

Compressed sensing has emerged as a promising nonlinear approach to address this sidelobe problem. By exploiting scene sparsity, CS enables superior reconstruction with suppressed sidelobes compared to linear methods [10]–[12]. However, its applicability is limited by signal sparsity and available measurements. To characterize these limits, theoretical frameworks such as the restricted isometry property (RIP) and phase transition theory have been established [13], [14].

In airborne random PRI radar, platform motion induces clutter spread, potentially compromising scenario sparsity [15].

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Furthermore, measurements are limited by the coherent processing interval. Therefore, a rigorous analysis of CS performance boundaries in this specific scenario is urgently needed. Existing analyses often assume generic matrices (e.g., Gaussian or row-orthogonal) [16], [17] and lack integration with realistic radar models, making it difficult to apply theoretical bounds to practical design guidelines.

To address this, this paper analyzes CS performance boundaries in random PRI radar via phase transition, which is more intuitive and computationally tractable than RIP. First, we empirically demonstrate that the structured random PRI matrix exhibits Gaussian-like phase transition behavior, justifying Gaussian-based theory for performance analysis. Second, we derive a concrete boundary condition with parameters explicitly linked to radar system and clutter geometry, providing a direct design tool. Finally, we validate this derived boundary through extensive simulations, confirming its effectiveness in predicting stable recovery transitions in both noiseless and noisy environments.

The remainder of this paper is organized as follows. Section II formulates the signal model. Section III analyzes the performance boundaries. Section IV presents numerical validations, and Section V concludes the paper.

Notation: In this paper, we use uppercase and lowercase boldface letters for matrices and vectors, respectively. The (m, n) -th (n -th) element of matrix $\mathbf{A}$ (vector $\mathbf{a}$) is written as $A_{m,n}$ (a_n). The operators $(\cdot)^H$ and $(\cdot)^T$ are the Hermitian transpose and transpose, respectively. $\lfloor \cdot \rfloor$, $\lceil \cdot \rceil$ are the floor and ceiling function. $\|\cdot\|_p$ represents the L_p -norm.

II. PROBLEM FORMULATION

This section formulates the signal model and CS framework for random PRI radar. We introduce the random PRI waveform and its echo signal model, then formulate scenario reconstruction as a CS problem via range-Doppler discretization, and adopt the least absolute shrinkage and selection operator (LASSO) method for scenario reconstruction.

A. Signal Model

Consider an airborne radar transmitting a coherent train of M pulses. The n -th pulse is transmitted at time $nT_r + \epsilon_n$,

where T_r is the nominal PRI and $\epsilon_n \sim U[-\xi/2, \xi/2]$ is the random jitter. The transmitted signal is

$$s_t(t) = \sum_{n=0}^{M-1} a(t - nT_r - \epsilon_n), \quad (1)$$

where $a(t)$ is the baseband waveform (e.g., a chirp). The received signal $s_r(t)$ comprises echoes from both targets and ground clutter, along with additive noise $n(t)$:

$$s_r(t) = \sum_{i \in \mathcal{I}} \alpha_i s_t(t - \tau_i) e^{j2\pi f_{d,i} t} + n(t), \quad (2)$$

where $\mathcal{I}$ indexes all scatterers (including K_t targets and N_c clutter points), and $\alpha_i, \tau_i, f_{d,i}$ denote the amplitude, delay, and Doppler frequency of the i -th scatterer, respectively.

B. Compressed Sensing Modeling

In this subsection, we construct a CS model to reconstruct the target scenario. By sampling the received signal $s_r(t)$ at time instants $\mathbf{t} = [0, T_s, \dots, (N-1)T_s]^T$, we obtain the measurement vector $\mathbf{y} = s_r(\mathbf{t}) \in \mathbb{C}^N$.

We define the range and Doppler steering vectors as $\mathbf{u}(\tau) = s_t(\mathbf{t} - \tau)$ and $\mathbf{v}(f) = e^{j2\pi f \mathbf{t}}$, respectively. By discretizing the region of interest into a grid of P delays $\tau_1, \dots, \tau_P$ and Q Doppler frequencies $f_1, \dots, f_Q$, we form the dictionary matrices:

$$\mathbf{U} = [\mathbf{u}(\tau_1), \dots, \mathbf{u}(\tau_P)], \quad \mathbf{V} = [\mathbf{v}(f_1), \dots, \mathbf{v}(f_Q)]. \quad (3)$$

The measurement matrix $\mathbf{A} \in \mathbb{C}^{N \times PQ}$ is then constructed as the Khatri-Rao product of the Doppler and range dictionaries:

$$\mathbf{A} = [\mathbf{V}^T \otimes \mathbf{U}]^T. \quad (4)$$

Consequently, the signal model is formulated as an underdetermined linear system:

$$\mathbf{y} = \mathbf{A}\mathbf{x} + \mathbf{n}, \quad (5)$$

where $\mathbf{x} \in \mathbb{C}^{PQ}$ is the sparse vector representing the scattering coefficients of targets and clutter, and $\mathbf{n}$ denotes the additive noise.

In random PRI radar, the unambiguous domain is significantly enlarged. To cover this enlarged area, we require $P > T_r B$ and $Q > M$, leading to $PQ > N$. Consequently, $\mathbf{A}$ is a fat matrix ($N < PQ$), making (5) an underdetermined system suitable for CS reconstruction.

C. LASSO Solution

Having established the CS model, we formulate an optimization problem to solve it. To recover the target scenario vector $\mathbf{x}$, we employ the LASSO estimator, which promotes sparsity via ℓ_1 -norm regularization:

$$\hat{\mathbf{x}} = \arg \min_{\mathbf{x}} \left\{ \frac{1}{2} \|\mathbf{y} - \mathbf{A}\mathbf{x}\|_2^2 + \lambda \|\mathbf{x}\|_1 \right\}, \quad (6)$$

where $\lambda > 0$ is a regularization parameter.

It is crucial to note that the solution $\hat{\mathbf{x}}$ reconstructs both targets and clutter scatterers simultaneously. Since clutter energy is concentrated in specific regions of the range-Doppler plane, it contributes significant non-zero entries to $\mathbf{x}$, thereby directly determining the effective sparsity level K of the scene. Consequently, the successful suppression of clutter sidelobes via CS is fundamentally achieved through the precise reconstruction of these dominant clutter components.

Admittedly, the dimension of $\mathbf{A}$ in wide-area scenarios incurs high computational complexity for CS recovery. Since this paper focuses on theoretical performance bounds, algorithmic acceleration is deferred to future work.

III. PERFORMANCE BOUND ANALYSIS

This section first investigates the CS performance boundaries by leveraging the phase transition theory. Then we derive the exact reconstruction condition by calculating the clutter sparsity level from the radar geometry. Finally, the analysis is extended to noisy scenarios to predict performance limits.

A. Phase Transition Phenomenon in CS

Phase transition theory characterizes the sharp performance boundary of CS recovery, dividing the parameter space into success and failure regions. We consider the exact reconstruction of the target scene $\mathbf{x}$ from noiseless measurements $\mathbf{y} = \mathbf{A}\mathbf{x}$ via standard ℓ_1 -minimization:

$$\hat{\mathbf{x}} = \arg \min_{\mathbf{x}} \|\mathbf{x}\|_1, \quad \text{s.t. } \mathbf{y} = \mathbf{A}\mathbf{x}. \quad (7)$$

For complex-valued Gaussian measurement matrices, the theoretical minimum number of measurements required for exact recovery is quantitatively given by [17]:

$$\varphi_2^c(s, d) := \frac{1}{2} \inf_{\tau \geq 0} \left\{ s(2 + \tau^2) + (d - s) \int_{\tau}^{\infty} (u - \tau)^2 u e^{-u^2/2} du \right\}, \quad (8)$$

where s denotes the sparsity level (number of non-zeros) of the vector $\mathbf{x}$, and d represents its total dimension. Consequently, the condition $N > \varphi_2^c(s, d)$ serves as the fundamental criterion for successful reconstruction.

Although the random PRI measurement matrix (constructed from Toeplitz and Fourier bases) deviates from the standard Gaussian assumption, empirical evidence suggests universality in phase transitions [18]. To verify this applicability, following

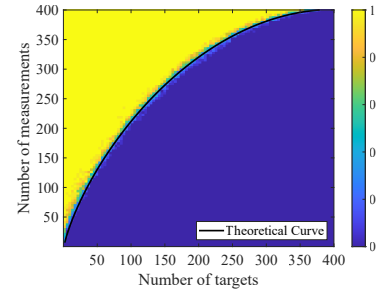


Fig. 1: Phase transitions of random PRI measurement matrices in the noiseless case, compared with the theoretical curve.

the experimental setting of Section IV-C we compare the empirical phase transition of the random PRI matrix with the theoretical curve in Fig. 1. The results demonstrate a consistent alignment, thereby validating the use of the Gaussian phase transition curve (8) to analyze the performance boundaries in random PRI scenarios.

B. Phase Transition Boundaries in Random PRI Scenarios

In the context of random PRI radar, the phase transition theory characterizes the exact reconstruction condition in the

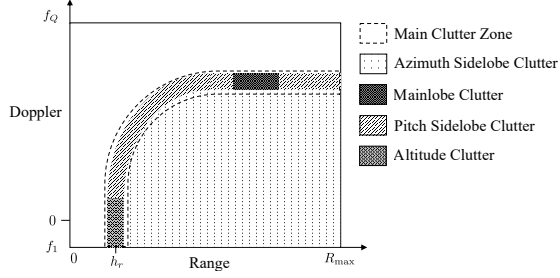


Fig. 2: Schematic diagram of the ground clutter distribution in the range-Doppler plane for airborne radar.

noiseless setting as

$$N > \varphi_2^c(K, N_{RD}), \quad (9)$$

where N is the number of measurements, N_{RD} denotes the dimension of the range-Doppler grid, and K represents the sparsity level (dominated by clutter). Crucially, since CS reconstructs both targets and clutter, K must account for the grid points occupied by both. This inequality establishes the theoretical minimum number of measurements required for exact reconstruction.

The parameters are calculated as follows. Assuming Nyquist sampling with respect to the bandwidth B , the total number of measurements is given by $N = M[T_r B]$. The grid dimension $N_{RD} = PQ$ represents the total number of grid points in the reconstructed range-Doppler map. Following (3), the number of grid points P and Q are determined by the region of interest, which is defined by the ambiguity orders n_r and n_v (i.e., $P \approx n_r[T_r B]$, $Q \approx n_v M$). It is worth noting that when selecting the surveillance region, all significant clutter returns should be included. If the selected region is clutter-free, the scenario inherently exhibits high sparsity, which is outside the scope of this boundary analysis.

The sparsity level K corresponds to the number of non-zero grid points occupied by clutter on the discretized range-Doppler plane. In airborne scenarios, clutter energy spans a wide area due to beam broadening. However, as illustrated in Fig. 2, we focus on quantifying the scatterers within the main clutter zone (mainlobe, pitch sidelobe, and altitude clutter), while neglecting the weak azimuth sidelobe clutter. To this end, we analyze the mapping from Doppler velocity to range using the idealized clutter ridge model defined by the radar altitude h_r , velocity v_r , and beam azimuth θ :

$$\frac{h_r^2}{r^2} + \frac{v^2}{(v_r \cos \theta)^2} = 1. \quad (10)$$

Based on this model, the idealized clutter range $R_c(v)$ for a given radial velocity v is determined by

$$R_c(v) = \min \left(R_{\max}, \sqrt{\frac{h_r^2 v_r^2 \cos^2 \theta}{v^2 \cos^2 \theta - v^2}} \right), \quad (11)$$

where $R_{\max}$ is the maximum range of the region of interest.

Accounting for grid discretization and practical clutter spread (e.g., due to antenna beamwidth), the number of occupied range bins $N_r(v_j)$ at Doppler bin v_j is calculated by:

$$N_r(v_j) = \left\lceil \frac{R_{\text{upper}}(v_j) - R_{\text{lower}}(v_j)}{\Delta R} \right\rceil, \quad (12)$$

where the range boundaries are given by:

$$\begin{aligned} R_{\text{upper}}(v_j) &= R_c(v_j - \Delta V/2) + \delta_R, \\ R_{\text{lower}}(v_j) &= R_c(v_j + \Delta V/2) - \delta_R. \end{aligned} \quad (13)$$

Here, ΔR , ΔV , δ_R represents the range resolution, velocity resolution and the range margin derived from the clutter spread. Note that for velocities outside the physical clutter spectrum (i.e., $|v_j| \geq v_r \cos \theta$), we set $N_r(v_j) = 0$. Finally, given that the scene sparsity is dominated by clutter, the total sparsity level is approximated as $K \approx \sum_{j=1}^Q N_r(v_j)$.

In conclusion, substituting the derived parameters N , N_{RD} , and K into the criterion (9) allows for an assessment of exact reconstruction feasibility. This framework provides radar engineers with a rigorous guideline for optimizing system design to guarantee accurate reconstruction of the target scenario in airborne environments with random PRI waveform.

C. Performance Assessment in Noisy Environments

The preceding analysis established phase transition boundaries for exact reconstruction under ideal noiseless conditions. To address practical applications, we extend the framework to noisy scenarios, aiming to determine whether the theoretical curves remain valid for characterizing the performance limits of the LASSO algorithm.

In the presence of noise, exact reconstruction is practically impossible, resulting in a deviation between the reconstructed signal and the ground truth. Consequently, the focus shifts to evaluating the estimation error. We hypothesize that the estimation performance under noisy conditions exhibits a phase transition behavior similar to the noiseless case, well-characterized by the Gaussian curve (8). To quantify this, we employ the Normalized Mean Squared Error (NMSE), $\text{NMSE} = \|\hat{x} - x\|_2^2 / \|x\|_2^2$. Although a rigorous derivation for structured matrices in noisy regimes is intractable, we assume the Gaussian curve remains a valid predictor for performance breakdown. This implies that satisfying $N > \varphi_2^c(K, N_{RD})$ is critical for maintaining low estimation error. The validity of this extension is empirically confirmed in Section IV.

IV. NUMERICAL EXPERIMENTS

Numerical simulations are conducted to validate the accuracy of the derived clutter zone model and to verify the effectiveness of the Gaussian phase transition curve in predicting performance boundaries under noisy conditions.

A. Setting

Consider an airborne radar platform moving along the y -axis with a constant velocity v_r at a fixed altitude h_r . The key parameters of the airborne radar scenario are given in Table I. The ground clutter is generated using the Morchin model [19] for hilly terrain, modeled on a denser grid to simulate off-grid clutter scenarios. Let α_t be the target coefficient and $\sigma_c^2 = \sum_l |\alpha_{c,l}|^2$ be the total clutter power. The Signal-to-Noise Ratio (SNR) and Clutter-to-Noise Ratio (CNR) are defined as:

$$\text{SNR} = 10 \log_{10} \left(\frac{G|\alpha_t|^2}{\sigma_n^2} \right), \quad \text{CNR} = 10 \log_{10} \left(\frac{G\sigma_c^2}{\sigma_n^2} \right), \quad (14)$$

TABLE I: Key Radar Simulation Parameters

Parameter	Value
Platform altitude	2000 m
Platform velocity	100 m/s
Array antenna elements	32×32
Beam azimuth angle	0°
Beam azimuth angle range	$[-10^\circ, 10^\circ]$
Beam pitch angle	-10°
Beam pitch angle range	$[-90^\circ, -2^\circ]$
Pulse number	100
Baseband signal	Chirp
Bandwidth	1 MHz
Carrier frequency	10 GHz
Reference PRI	$50 \mu s$
Pulse width	$5 \mu s$
Random PRI jitter	$0.5T_r$

where $G = T_p f_s M$ is the processing gain and σ_n^2 is the noise variance. The primary performance metric is the NMSE between the estimated and true scene vectors.

B. Clutter Zone Verification

To verify the accuracy of the derived clutter zone model, we simulate a clutter-dominated scenario with a CNR of 30 dB. The radar transmits 100 chirp pulses, and the region of interest is defined to encompass all significant clutter returns, discretized according to the inherent range and Doppler resolutions. The target scenario is reconstructed by solving the LASSO problem via the FISTA algorithm [20].

The resulting range-Doppler map is shown in Fig. 3. The

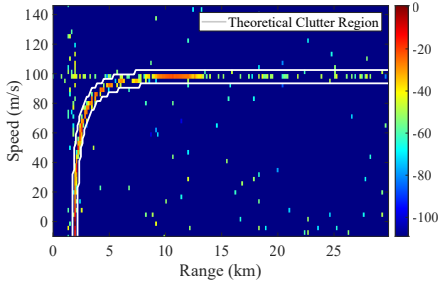


Fig. 3: Range-Doppler map recovered by LASSO. The white contour outlines the theoretical main clutter region.

theoretical main clutter region, calculated using the proposed model with appropriate spread parameters, is superimposed as a white contour. It is evident that the dominant clutter energy is tightly confined within the predicted boundary, while the energy outside this region is negligible. This alignment validates the accuracy of our clutter zone model, confirming that quantifying the grid occupancy within this zone provides a reliable estimation of the sparsity level K .

C. Phase Transition and Performance Bounds

First, we verify the theoretical minimum measurement requirement in a realistic clutter scenario. In this experiment, clutter scatterer locations are aligned with the grid points to enable exact evaluation, while their amplitudes are set according to the physical clutter simulation to reflect a realistic dynamic range. For computational efficiency, we set pulse number to 32 for phase transition analysis. The target scenario contains only clutter components, as they dominate the sparsity level. We evaluate the recovery performance using the NMSE, and

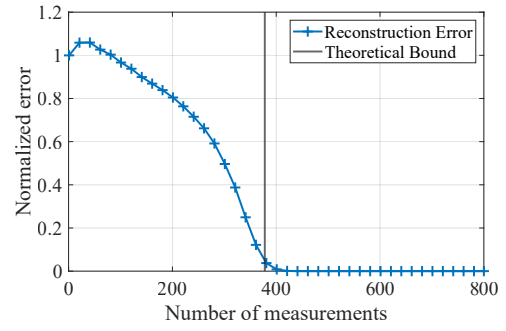


Fig. 4: NMSE of reconstruction under varying measurement numbers.

the number of measurements is varied by randomly selecting rows from the full measurement matrix. The results in Fig. 4 show that when measurements exceed the theoretical minimum (black line), the error stabilizes at a low level.

Next, to comprehensively analyze phase transitions across varying sparsity levels, we adopt a reduced problem size of $N_{RD} = 400$ (representing a scenario with four-fold range ambiguity and 10 Doppler bins) to mitigate the computational cost of large-scale simulations, as phase transitions are observable even in small-scale systems [18].

We extend the analysis to noisy scenarios using the NMSE of the LASSO estimator, as shown in Fig. 5. Due to the

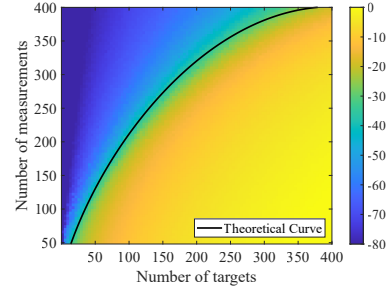


Fig. 5: Phase transition of NMSE for the LASSO estimator.

presence of noise, the phase transition boundary becomes less sharp compared to the noiseless case. Nevertheless, the success and failure phases remain distinct, with the NMSE below -50 dB and above -20 dB respectively, and their boundary aligns with the theoretical curve derived from the noiseless case. These results demonstrate that even in the presence of noise, the estimation performance metrics exhibit similar phase transition trends to the noiseless case. This confirms that the derived boundary serves as a valid predictor for performance limits even in practical noisy environments.

V. CONCLUSION

This paper established CS performance boundaries for airborne random PRI radar using phase transition theory. We verified the Gaussian universality of the random PRI matrix and derived a concrete reconstruction condition, $N > \varphi_2^c(K, N_{RD})$, explicitly linking theoretical parameters to radar geometry. Simulations confirm this boundary effectively predicts stable recovery boundaries in both noiseless and noisy environments, providing theoretical foundation and practical guidelines for waveform design in complex clutter scenarios.

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